Chemical Control of Respiration

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DIKIOHS,DUHS.

LEARNING OBJECTIVES

At the end of lecture the student should be able to

- Describe about the mechanisms of control of breathing.
- Enlist the changes in chemical constituents of blood which stimulate chemical receptors.
- Explain the mechanism of action of central chemoreceptors.
- Explain the mechanism of action of peripheral chemoreceptors.

Chemical control of respiration

- Rate & depth of respiration is also controlled by certain "receptors" lies in brain and outside the brain, which are sensitive to chemical, and whenever the chemical composition of blood is altered these receptors get stimulated.
- Chemical mechanism of regulation of respiration is operated through the chemoreceptors.
- **Chemoreceptors** are the sensory nerve endings, which give response to changes in chemical constituents of blood.

Types of Chemoreceptors

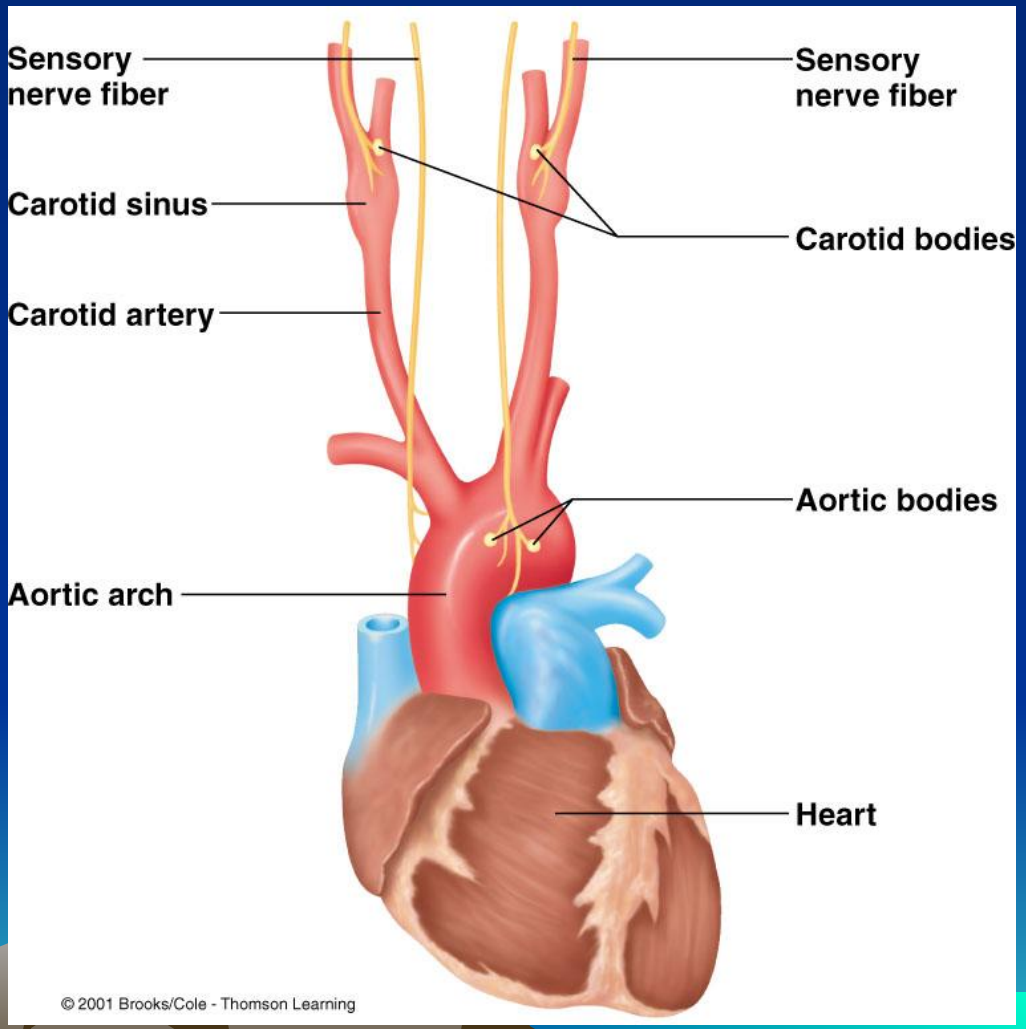
Chemoreceptors are classified into two groups:

1. Central chemoreceptors 2. Peripheral chemoreceptors

Chemical regulation of respiration

1. Peripheral chemoreceptor
   A. Aortic Bodies
   B. Carotid Bodies

2. Central Chemoreceptor



Chemical control of respiration

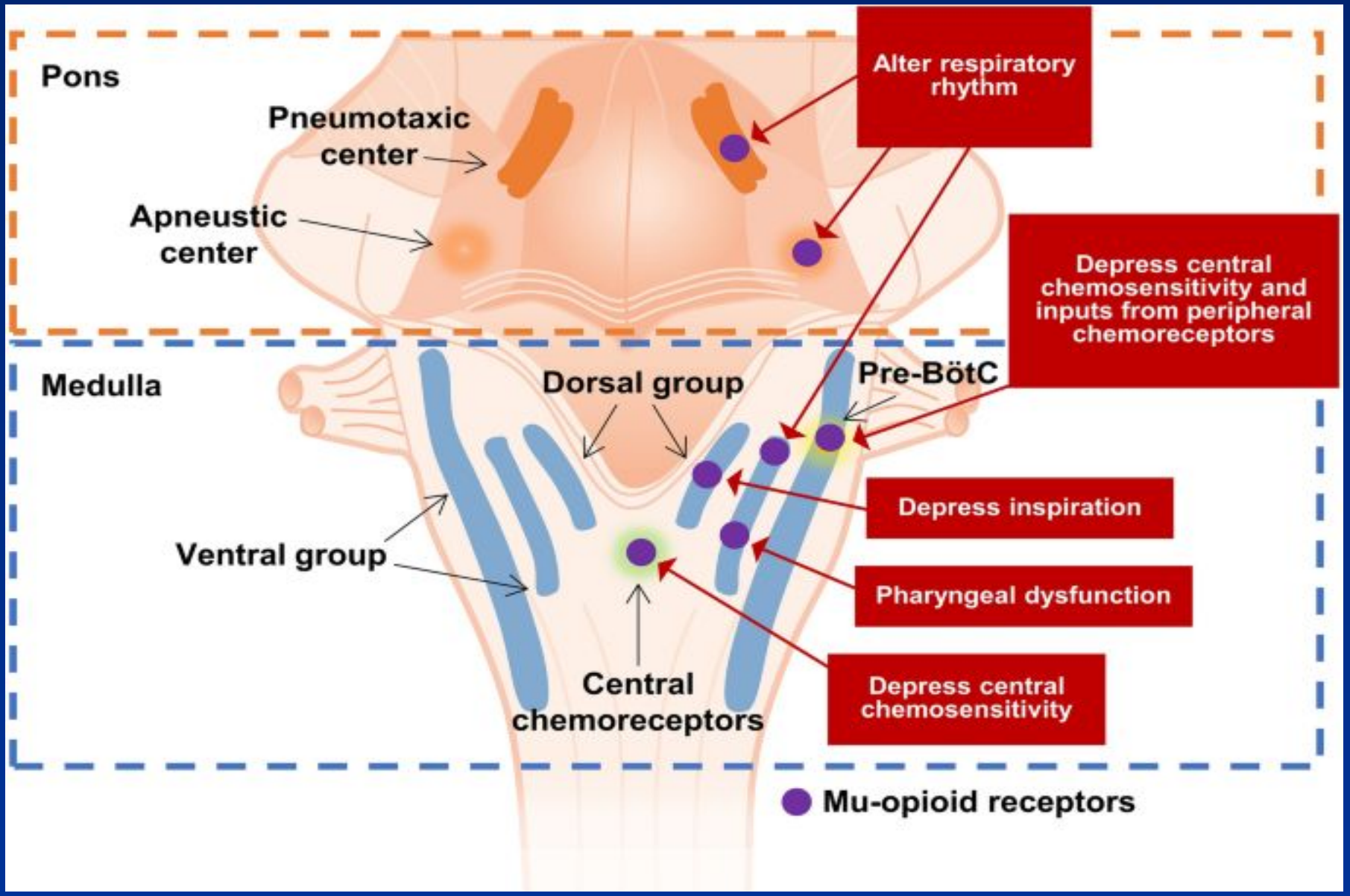
- **Changes in Chemical Constituents of Blood which Stimulate Chemoreceptors**

1. Hypoxia (decreased pO2)
2. Hypercapnia (increased pCO2)
3. Increased hydrogen ion concentration.

1.Central Chemoreceptor

- The central chemoreceptors, located in the brain stem, are the most important for the minute-to-minute control of breathing.
- These chemoreceptors are located on the ventral surface of the medulla, near the point of exit of the glossopharyngeal (CN IX) and vagus(CN X) nerves and only a short distance from the medullary inspiratory center.
- Thus, central chemoreceptors communicate directly with the inspiratory center.
- This area is known as chemosensitive area and the neurons are called chemoreceptors.Chemoreceptors are in close contact with blood and cerebrospinal fluid.

Location of central chemoreceptors



Mechanism of Action

- Central chemoreceptors are connected with respiratory centers, particularly the dorsal respiratory group of neurons through synapses. These chemoreceptors act slowly but effectively & are responsible for 70% to 80% of increased ventilation through chemical regulatory mechanism.
- Main stimulant for central chemoreceptors is the increased hydrogen ion concentration. However, if hydrogen ion concentration increases in the blood, it cannot stimulate the central chemoreceptors because, the hydrogen ions from blood cannot cross the blood brain barrier and blood-cerebrospinal fluid barrier.

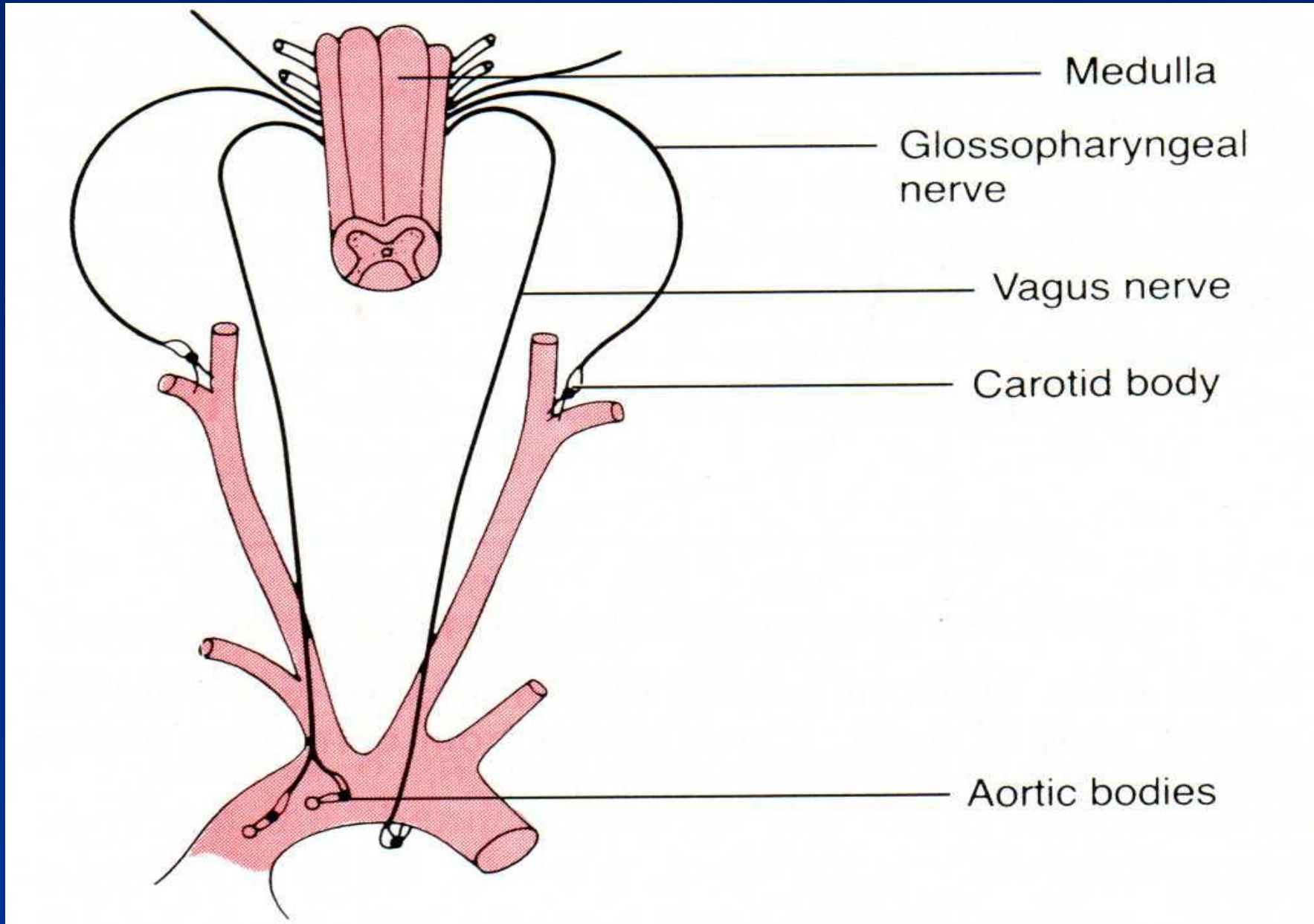
- On the other hand, if carbon dioxide increases in the blood, it can easily cross the blood-brain barrier and Blood cerebrospinal fluid barrier and enter the interstitial fluid of brain or the cerebrospinal fluid. There, the carbon dioxide combines with water to form carbonic acid. Since carbonic acid is unstable, it immediately dissociates into hydrogen ion and bicarbonate ion.

$CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3$

- Hydrogen ions stimulate the central chemoreceptors. From chemoreceptors, the excitatory impulses are sent to dorsal respiratory group of neurons, resulting in increased ventilation. Because of this, excess carbon dioxide is washed out and respiration is brought back to normal.

2- Peripheral chemoreceptors

- Mainly *the carotid and aortic bodies*.
- *Aortic bodies*. Lies in the walls of aortic arch.
- The Carotid Bodies: Lies at the bifurcation of common carotid artery.
- Information about arterial PO_2, PCO_2, and pH is relayed to the medullary inspiratory center via CN IX and CN X, which orchestrates an appropriate change in breathing rate.

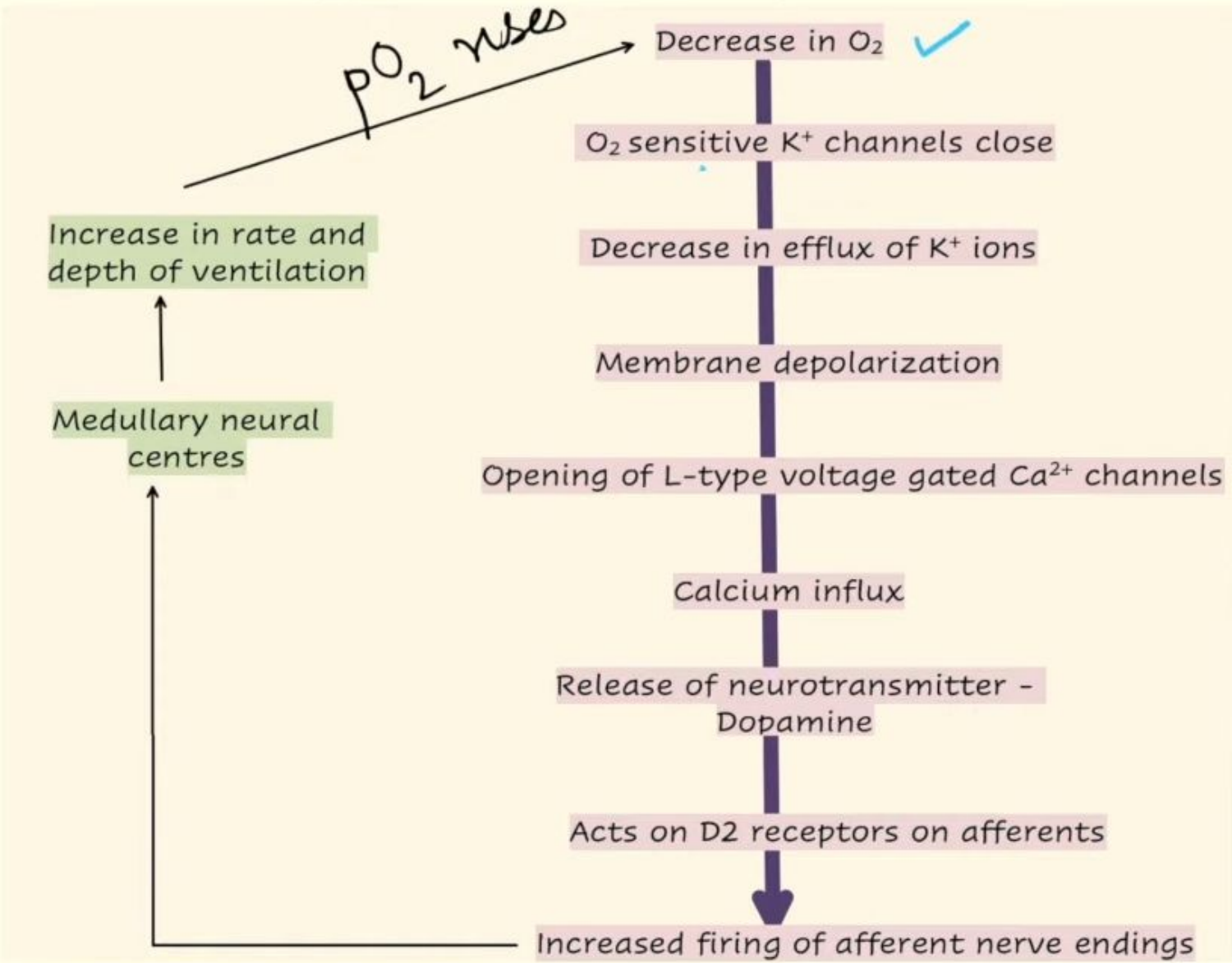


Respiratory control by the carotid and aortic bodies.

Mechanism of Action

- **Hypoxia is the most potent stimulant for peripheral chemoreceptors. It is because of the presence of oxygen sensitive potassium channels in the glomus cells of peripheral chemoreceptors.**
- **Hypoxia causes closure of oxygen sensitive potassium channels and prevents potassium efflux. This leads to depolarization of glomus cells (receptor potential) and generation of action potentials in nerve ending. These impulses pass through aortic and Hering nerves and excite the dorsal group of neurons, which send excitatory impulses to respiratory muscles, resulting in increased ventilation.**

Peripheral chemoreceptors

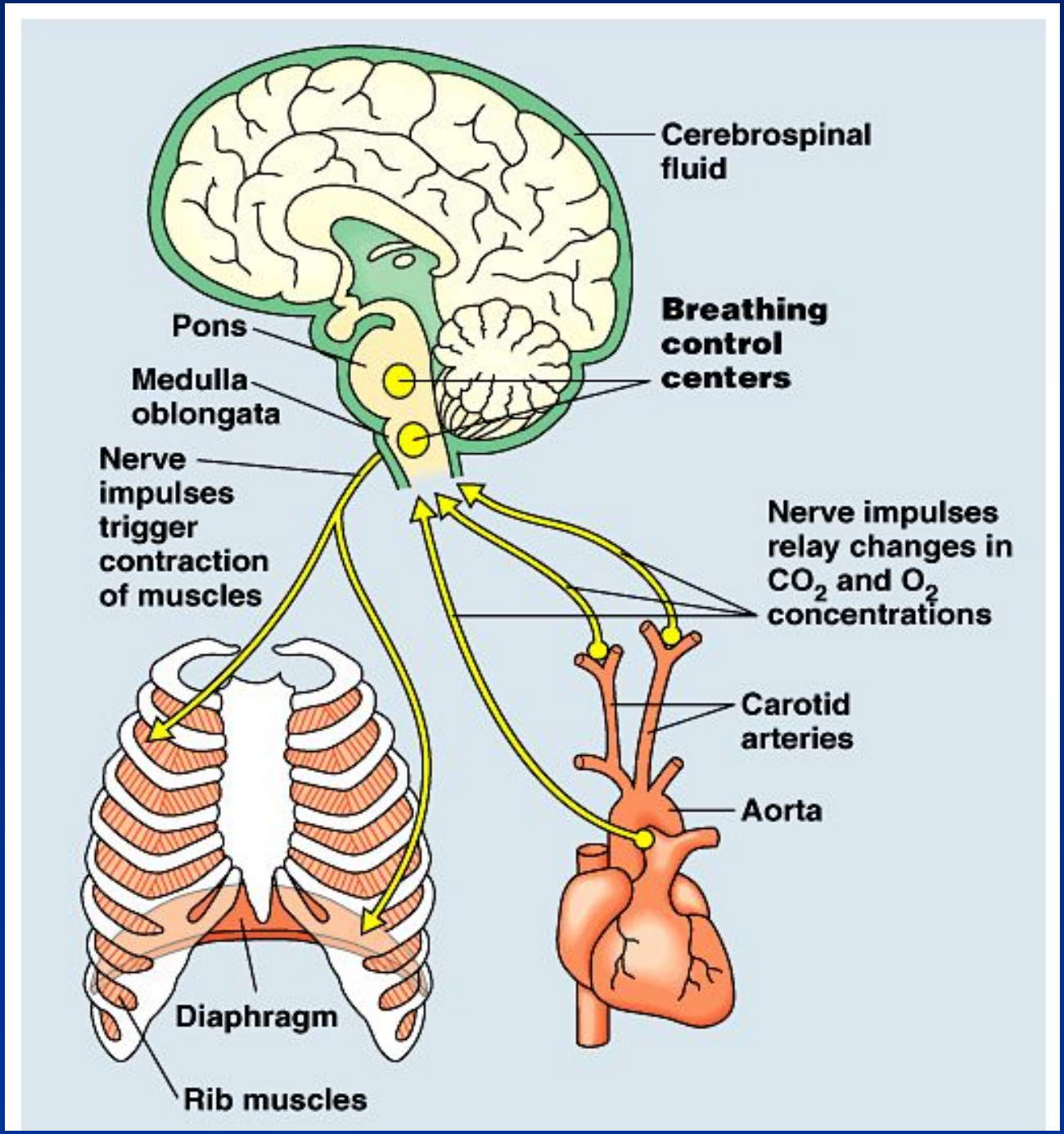


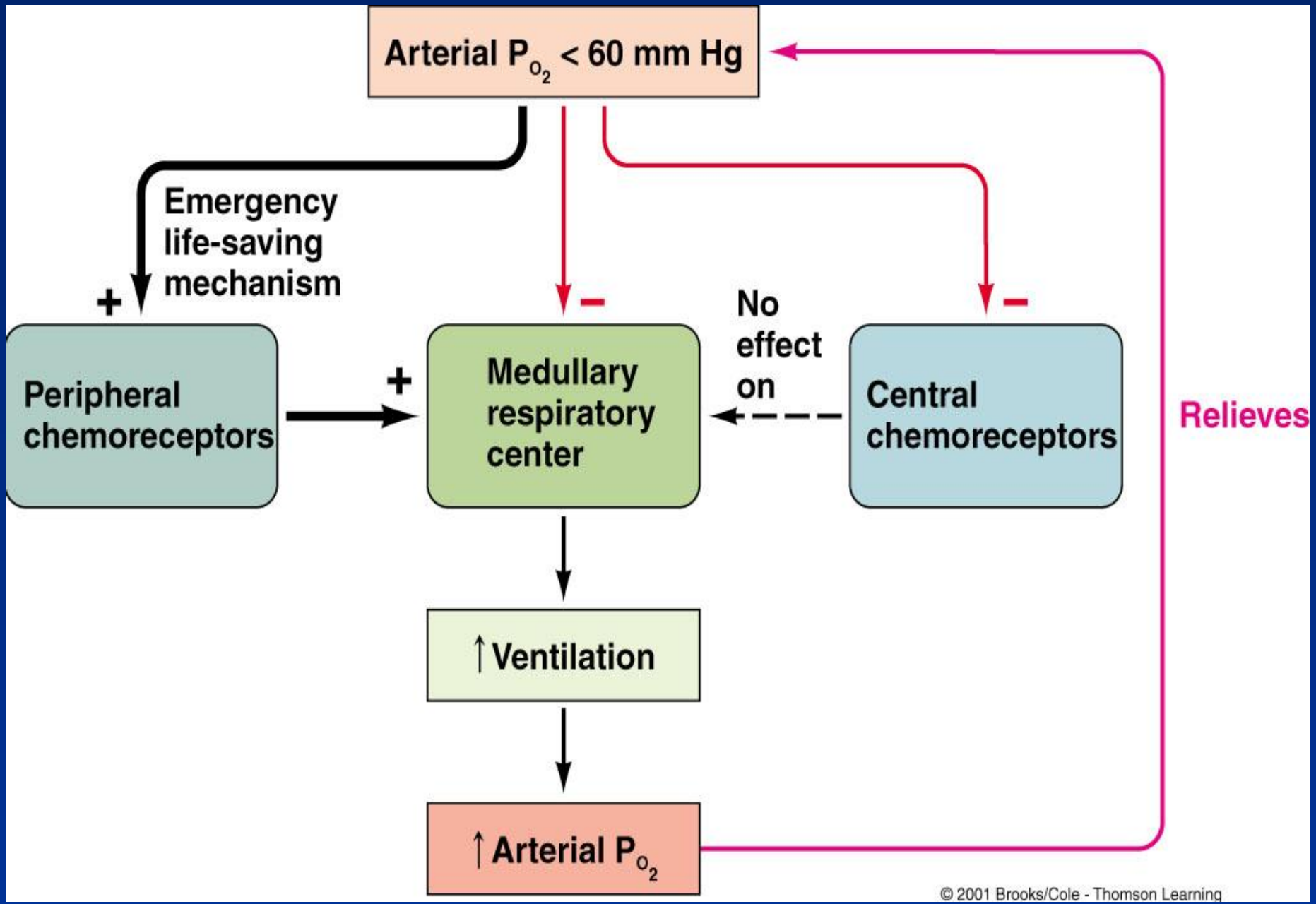
Decrease in arterial PO2

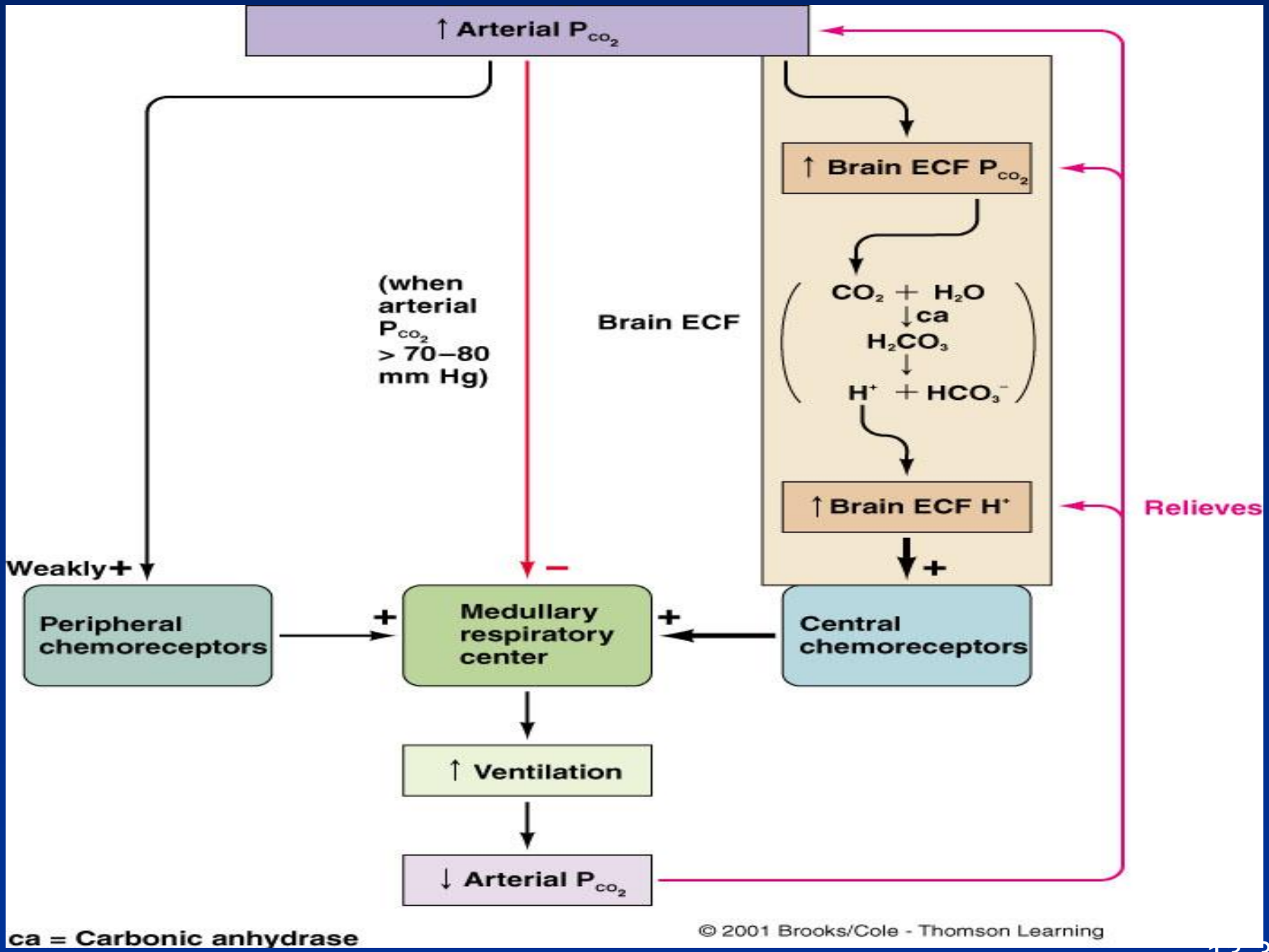
- The most important responsibility of the peripheral chemoreceptors is to detect changes in arterial PO2.however, the peripheral chemoreceptors are relatively insensitive to changes in PO2:
- They respond when PO2 decreases to less than 60 mm Hg. Thus, if arterial PO2 is between 100 mm Hg and 60 mm Hg, the breathing rate is virtually constant. However, if arterial PO2 is less than 60 mm Hg, the breathing rate increases in a very steep and linear fashion.
- In this range of PO2, chemoreceptors are exquisitely sensitive to O2; in fact, they respond so rapidly that the firing rate of the sensory neurons may change during a single breathing cycle.

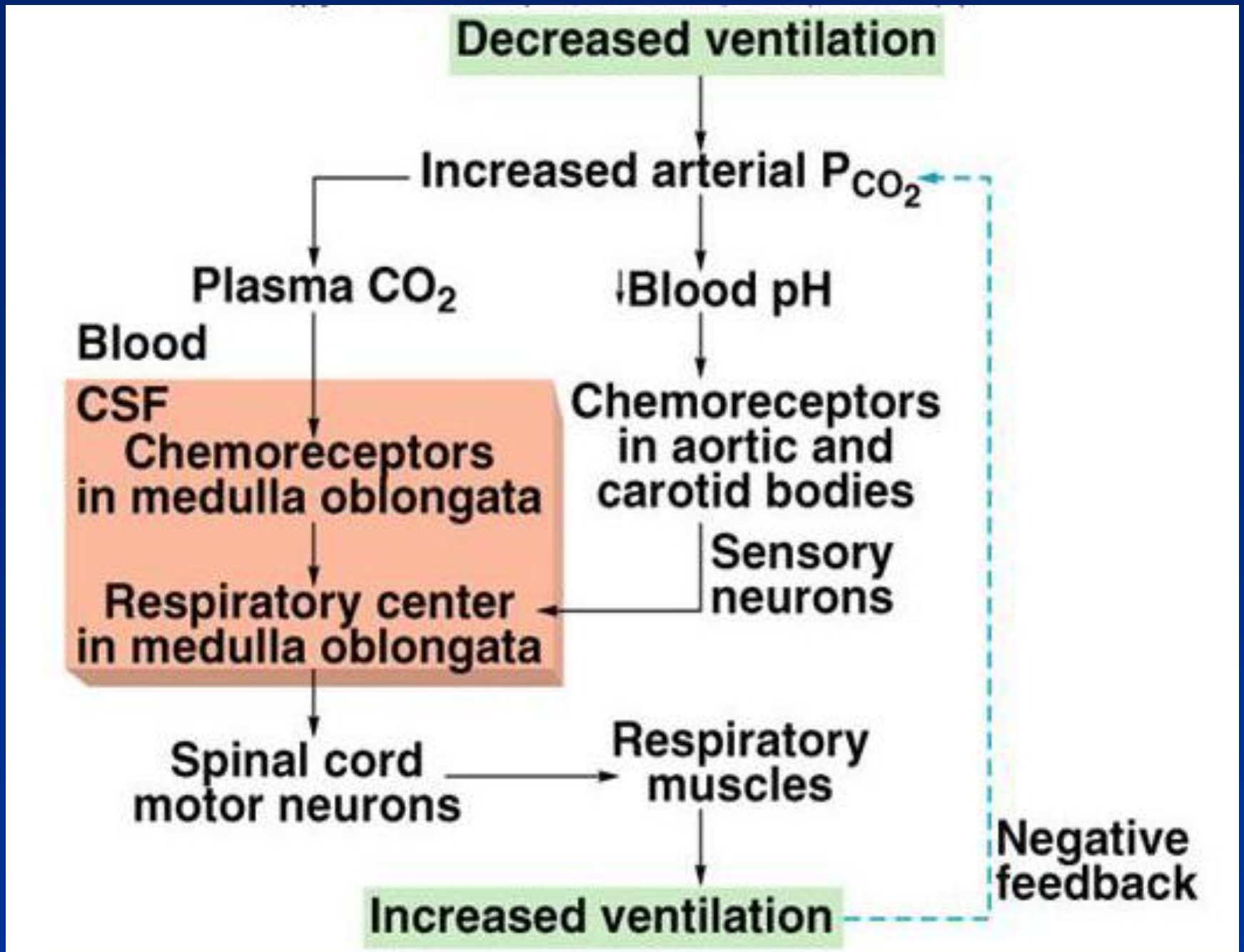
- In addition to hypoxia, peripheral chemoreceptors are also stimulated by hypercapnia and increased hydrogen ion concentration. However, the sensitivity of peripheral chemoreceptors to hypercapnia and increased hydrogen ion concentration is mild.

Control of breathing; gas level detectors




Arterial P_{O_2} < 60 mm Hg
Emergency life-saving mechanism
+
Peripheral chemoreceptors
+
Medullary respiratory center
–
No effect on
Central chemoreceptors
–
Relieves
↑ Ventilation
↑ Arterial P_{O_2}
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↑ Arterial P_{CO_2}
↑ Brain ECF P_{CO_2}
$CO_2 + H_2O$
↓ca
H_2CO_3
↓
$H^+ + HCO_3^-$
↑ Brain ECF H^+
(when arterial P_{CO_2} > 70–80 mm Hg)
Brain ECF
Relieves
Weakly +
−
+
+
+
Peripheral chemoreceptors
Medullary respiratory center
Central chemoreceptors
↑ Ventilation
↓ Arterial P_{CO_2}
ca = Carbonic anhydrase
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Decreased ventilation
Increased arterial P_{CO_2}
Plasma CO_2
Blood pH
Blood
CSF
Chemoreceptors in medulla oblongata
Chemoreceptors in aortic and carotid bodies
Sensory neurons
Respiratory center in medulla oblongata
Spinal cord motor neurons
Respiratory muscles
Increased ventilation
Negative feedback

Mechanical Control of Breathing:

- Sensory receptors in the lung and airways are stimulated by irritation of the mucosa and changes in the distending pressure.
- Afferent (sensory) neurons travel up the vagus to the brainstem areas controlling respiration.

Three types of pulmonary receptors:

1. Stretch receptors (regulatory)
2. Irritant receptors (protective)
3. C-fibers (protective)

Lung stretch receptors

- Hering Breuer reflex is a protective reflex that restricts inspiration & prevents overstretching of lung tissues. It is initiated by the stimulation of stretch receptors of air passage.
- Stretch receptors are the receptors which give response to stretch of the tissues. These receptors are situated on the wall of the bronchi and bronchioles.
- Expansion of lungs during inspiration stimulates the stretch receptors. Impulses from stretch receptors reach the dorsal group neurons via vagal afferent fibers and inhibit them. So, inspiration stops & expiration starts, thereby preventing overstretching of lung tissues.
- However, Hering Breuer reflex does not operate during quiet breathing. It operates, only when the tidal volume increases beyond 1,000 mL.

Joint and muscle receptors

- Mechanoreceptors located in the joints and muscles detect the movement of limbs and instruct the inspiratory center to increase the breathing rate.
- Information from the joints and muscles is important in the early (anticipatory) ventilatory response to exercise.

J receptors

- Juxtacapillary (J) receptors are located in the alveolar walls and, therefore, are near the capillaries. Engorgement of pulmonary capillaries with blood and increases in interstitial fluid volume may activate these receptors and produce an increase in the breathing rate. For example, in leftsided heart failure, blood “backs up” in the pulmonary circulation, and J receptors mediate a change in breathing pattern, including rapid shallow breathing and dyspnea (difficulty in breathing).

Irritant receptors

- Irritant receptors for noxious chemicals and particles are located between epithelial cells lining the airways.
- Information from these receptors travels to the medulla via CN X and causes a reflex constriction of bronchial smooth muscle and an increase in breathing rate.

VIDEO LINK

- https://youtu.be/zjSBtRjgFjE?si=XdhrhzsyjFzLh7h